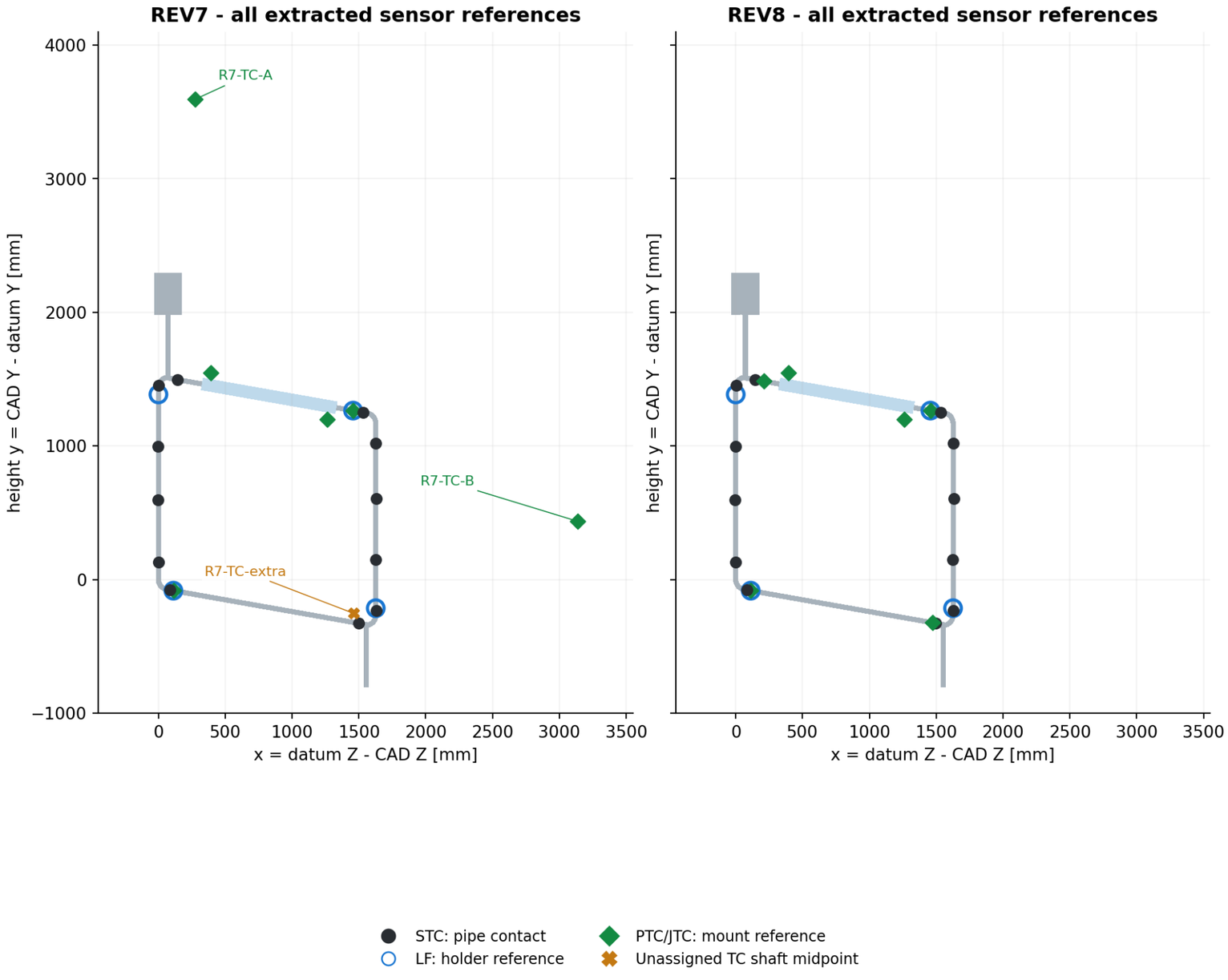
VCU Rev7 vs Rev8: direct CAD comparison

Direct STEP extraction | Coordinates and lengths in mm

Full CAD extent reveals the Rev7 off-loop TC placements



Main-loop geometry agrees. All eight extracted corner tangents have identical raw CAD XYZ coordinates. The straight lengths and bend dimensions also agree.

20 sensor references agree within 0.001 mm: 12 surface contacts, four LF mounts and four TC mounts. The largest computed difference is 1.83e-10 mm, numerical noise at this scale.

Rev7 has two remote TC mounts. R7-TC-A and R7-TC-B lie far outside the loop; Rev8 instead has PTC-01 and PTC-04 on the loop. A unique old-to-new identity is not encoded, so these remain unmatched.

Rev7 has one additional modeled TC shaft inside TC_30deg. Its midpoint is shown in amber; both shaft endpoints are retained. It has no established Rev8 counterpart.

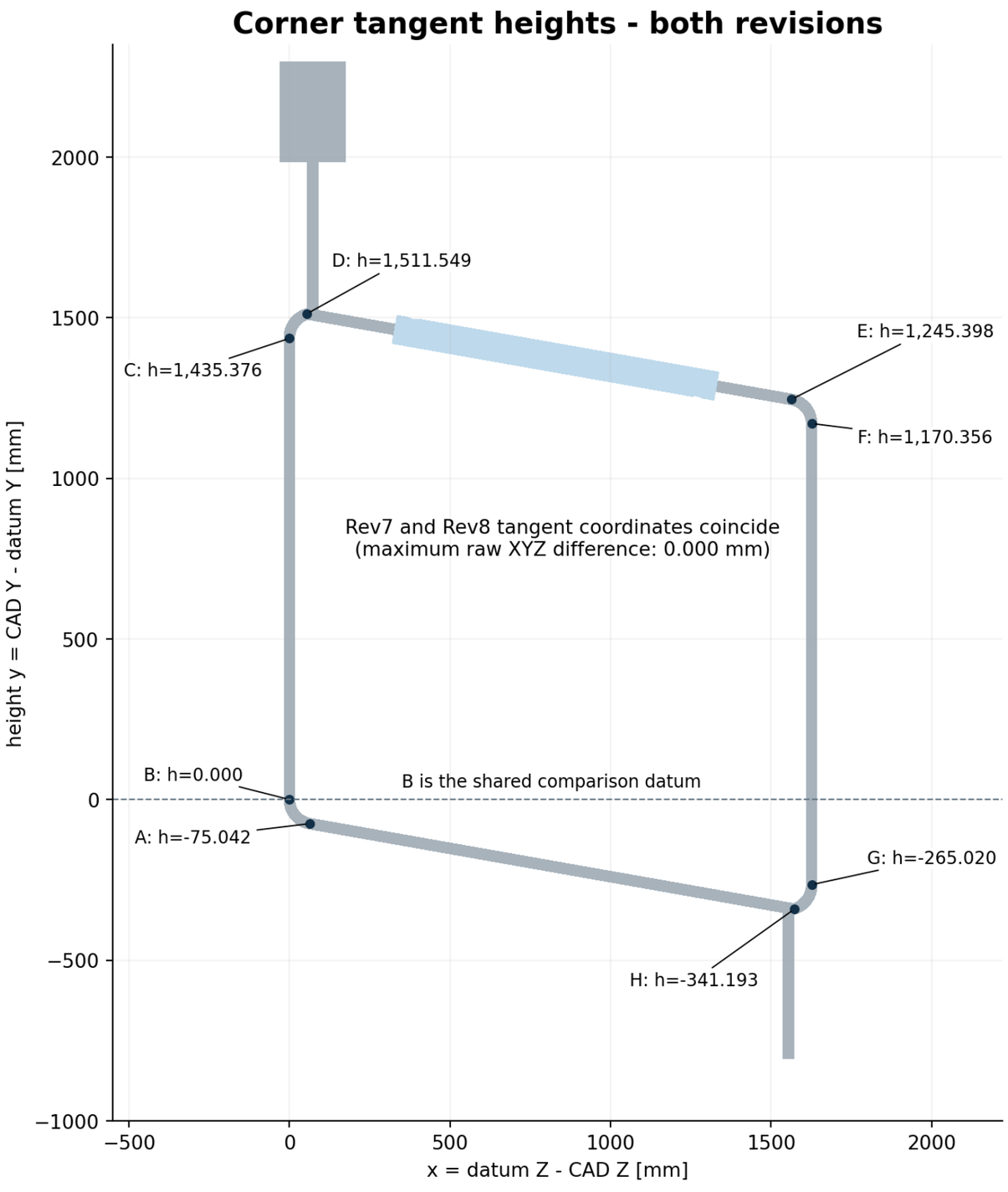
CAD imperfections are retained. Both models have a 0.272 mm offset at the cooler axis joint. Two surface-TC bodies have small nonzero gaps; maximum 0.092 mm.

This report compares the newly supplied **Rev7** file with **Rev8(1)**. It does not reuse the earlier unversioned STEP comparison or prior spreadsheet coordinates.

Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

Corner elevations and signed segment dimensions

Direct STEP extraction | Coordinates and lengths in mm



Shared datum B = lower tangent of the upcomer straight.
CAD XYZ = (629.806632391, 852.124195141, 3306.139984198) mm in both files.
x = datum Z - CAD Z; height y = CAD Y - datum Y; normal = CAD X - datum X.

Segment / direction	dx	dy	Length	Basis
heater H-A	-1,509.415	266.151	1,532.700	axis length
lower left bend A-B	-62.968	75.042	106.395	arc
upcomer B-C	-0.000	1,435.376	1,435.376	axis length
upper left bend C-D	53.821	76.173	93.268	chord only
cooler D-E	1,509.415	-266.151	1,532.700	axis length
upper right bend E-F	62.968	-75.042	106.395	arc
downcomer F-G	0.000	-1,435.376	1,435.376	axis length
lower right bend G-H	-53.821	-76.173	93.268	chord only

dx and dy follow heater -> upcomer -> cooler -> downcomer. Length is positive. For the two non-analytical joined corners, the displayed length is the endpoint chord, not an invented arc.

Exact lower-left and upper-right bends: R = 76.200 mm, angle = 80.000 deg, arc = 106.395 mm. The cooler length is the sum of two cylinder-axis pieces; their 0.272 mm offset joint is recorded separately.

The height range of the eight tangents is 1,852.741 mm. This is neither the tank-inclusive envelope nor an assumed buoyancy head. The prior A0 used CAD Y = 587.104 mm; to recover that prior height, add 265.020195 mm to the heights here.

Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

All corner tangent coordinates: both CAD files

Direct STEP extraction | Coordinates and lengths in mm

Rev	Tag	Corner / tangent	CAD X	CAD Y	CAD Z	Drawing x	Height y	Normal
rev7	A	lower left / heater outlet	629.807	777.082	3,243.172	62.968	-75.042	0.000
rev8	A	lower left / heater outlet	629.807	777.082	3,243.172	62.968	-75.042	0.000
rev7	B	lower left / upcomer inlet	629.807	852.124	3,306.140	-0.000	0.000	0.000
rev8	B	lower left / upcomer inlet	629.807	852.124	3,306.140	-0.000	0.000	0.000
rev7	C	upper left / upcomer outlet	629.807	2,287.500	3,306.140	-0.000	1,435.376	0.000
rev8	C	upper left / upcomer outlet	629.807	2,287.500	3,306.140	-0.000	1,435.376	0.000
rev7	D	upper left / cooler inlet	629.807	2,363.673	3,252.319	53.821	1,511.549	0.000
rev8	D	upper left / cooler inlet	629.807	2,363.673	3,252.319	53.821	1,511.549	0.000
rev7	E	upper right / cooler outlet	630.078	2,097.522	1,742.904	1,563.236	1,245.398	0.272
rev8	E	upper right / cooler outlet	630.078	2,097.522	1,742.904	1,563.236	1,245.398	0.272
rev7	F	upper right / downcomer inlet	630.067	2,022.480	1,679.936	1,626.204	1,170.356	0.261
rev8	F	upper right / downcomer inlet	630.067	2,022.480	1,679.936	1,626.204	1,170.356	0.261
rev7	G	lower right / downcomer outlet	629.821	587.104	1,679.936	1,626.204	-265.020	0.015
rev8	G	lower right / downcomer outlet	629.821	587.104	1,679.936	1,626.204	-265.020	0.015
rev7	H	lower right / heater inlet	629.807	510.931	1,733.757	1,572.383	-341.193	-0.000
rev8	H	lower right / heater inlet	629.807	510.931	1,733.757	1,572.383	-341.193	-0.000

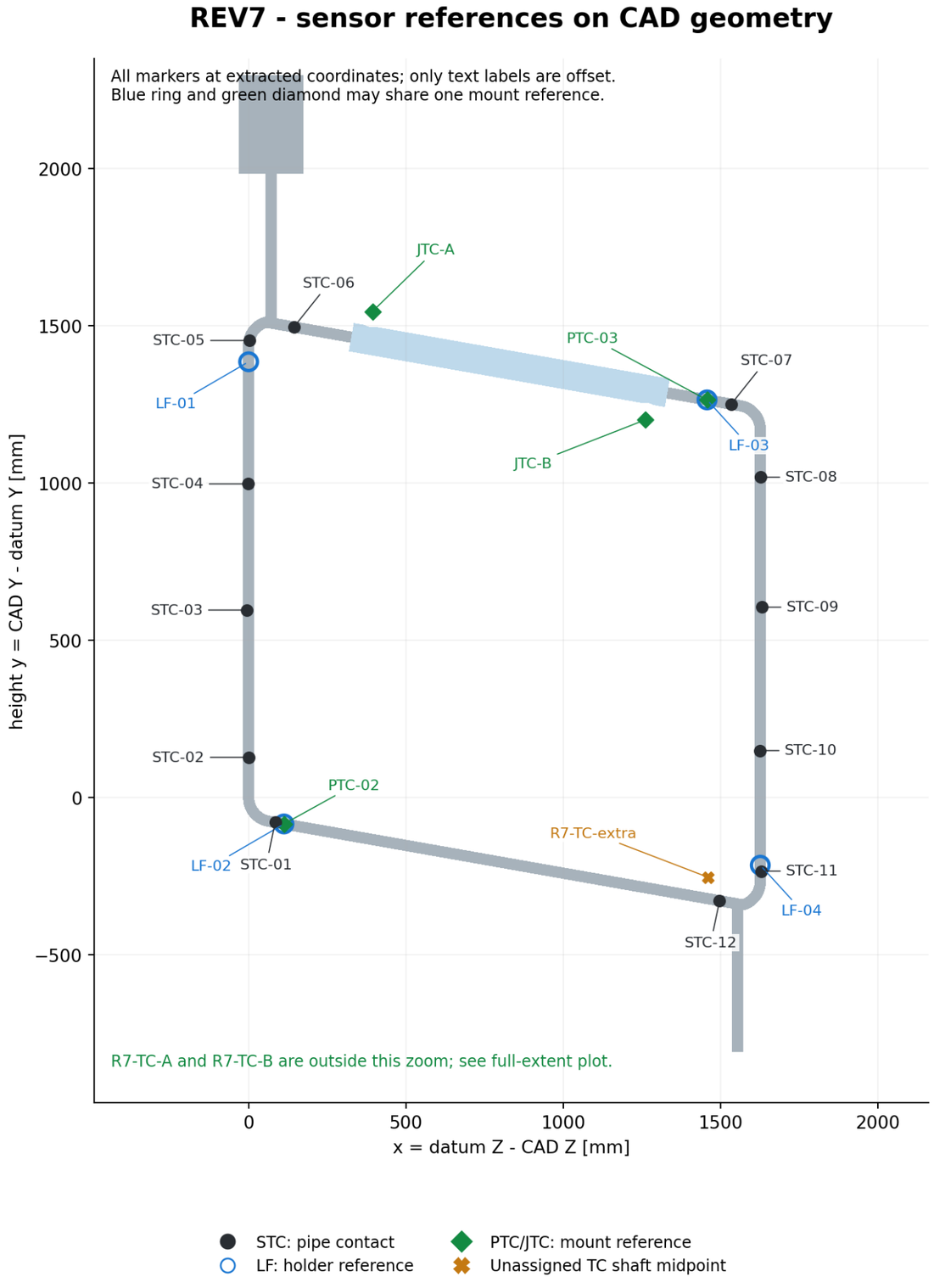
A/H: heater endpoints. B/C: upcomer endpoints. D/E: cooler endpoints. F/G: downcomer endpoints. Each curved corner has two tangent points; no single imaginary bend intersection is substituted.

Maximum Rev8-minus-Rev7 corner difference: 0.000 mm. Full-precision values, individual CAD face paths and signed delta columns are provided in corners.csv, corner_comparison.csv and the workbook.

Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

REV7 sensor map: actual CAD placements

Direct STEP extraction | Coordinates and lengths in mm



Black circles: closest point on the pipe solid to the modeled surface thermocouple. The physical contact and measured CAD gap are retained separately in the raw extraction.

Blue rings: fiber-holder mounting reference. A holder does not define the spatial extent of a Luna sensing fiber.

Green diamonds: TC mounting-body centroid. PTC-02/LF-02 and PTC-03/LF-03 share a combined assembly reference. That does not establish coincident sensing elements.

JTC-A and JTC-B: supplemental labels for mounts near jacket ports, separate from the main loop. Their service is inferred from geometry, not a named CAD channel.

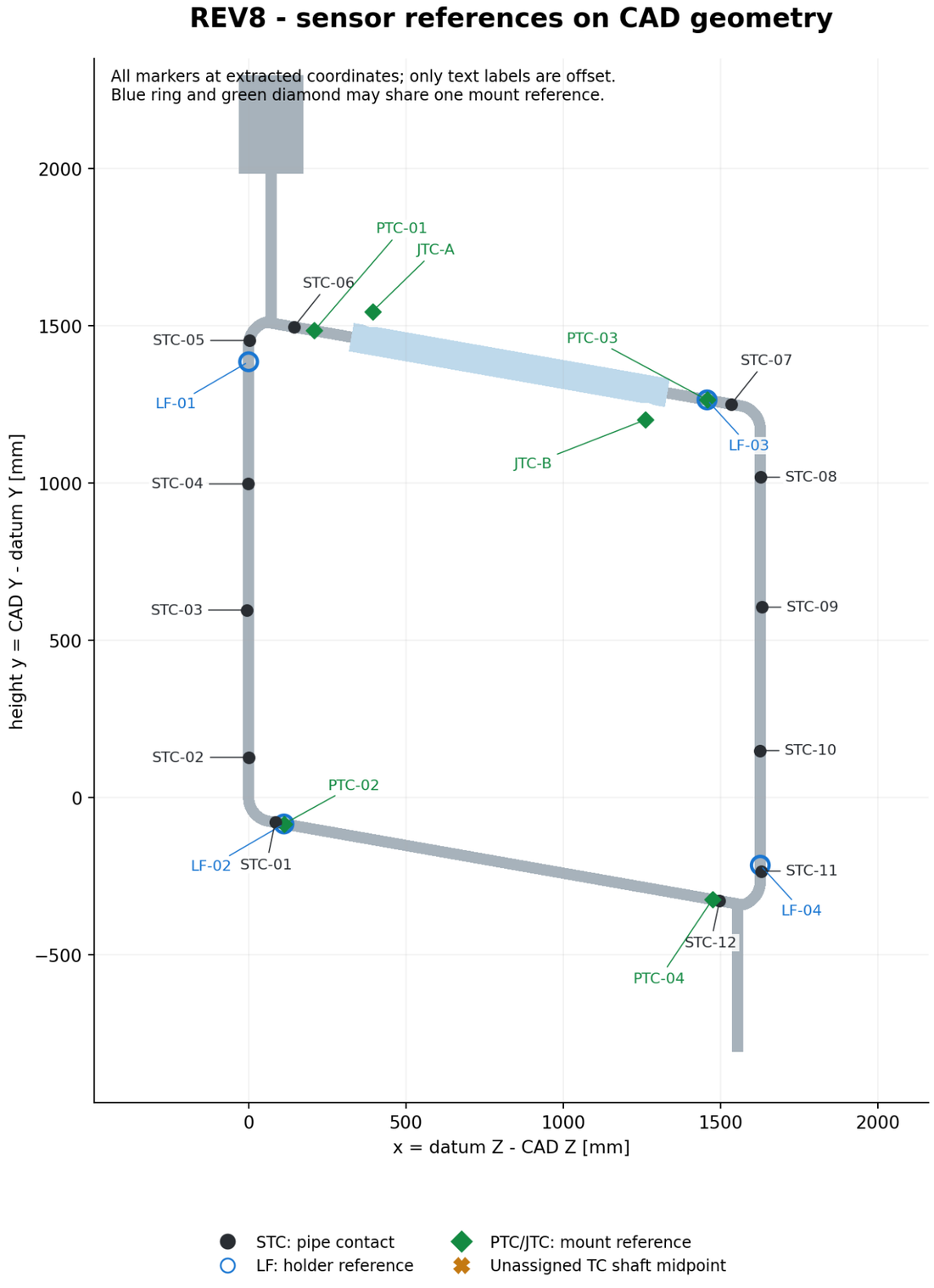
Coordinates are not snapped to the pipe. Only label text is offset. The additional green diamonds outside the pipe represent the actual extracted mounting references.

STC/LF/PTC numbering follows collaborator-schema loop regions as a drawing crosswalk. CAD paths are the authoritative record IDs; acquisition-channel identity remains unverified.

Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

REV8 sensor map: actual CAD placements

Direct STEP extraction | Coordinates and lengths in mm



Black circles: closest point on the pipe solid to the modeled surface thermocouple. The physical contact and measured CAD gap are retained separately in the raw extraction.

Blue rings: fiber-holder mounting reference. A holder does not define the spatial extent of a Luna sensing fiber.

Green diamonds: TC mounting-body centroid. PTC-02/LF-02 and PTC-03/LF-03 share a combined assembly reference. That does not establish coincident sensing elements.

JTC-A and JTC-B: supplemental labels for mounts near jacket ports, separate from the main loop. Their service is inferred from geometry, not a named CAD channel.

Coordinates are not snapped to the pipe. Only label text is offset. The additional green diamonds outside the pipe represent the actual extracted mounting references.

STC/LF/PTC numbering follows collaborator-schema loop regions as a drawing crosswalk. CAD paths are the authoritative record IDs; acquisition-channel identity remains unverified.

Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

REV7 sensor coordinate schedule

Direct STEP extraction | Coordinates and lengths in mm

ID	Type	CAD X	CAD Y	CAD Z	Drawing x	Height y	Normal	STEP path below root
JTC-A	PTC	630.458	2,397.359	2,912.585	393.555	1,545.235	0.652	NAUO11
JTC-B	PTC	628.995	2,053.093	2,044.757	1,261.383	1,200.969	-0.812	NAUO25
LF-01	LF	629.807	2,237.407	3,306.140	0.000	1,385.283	0.000	NAUO1/NAUO56
LF-02	LF	629.786	768.348	3,193.905	112.235	-83.776	-0.021	NAUO9
LF-03	LF	630.052	2,116.226	1,849.222	1,456.918	1,264.102	0.245	NAUO7
LF-04	LF	629.830	637.197	1,679.936	1,626.204	-214.927	0.023	NAUO2/NAUO56
PTC-02	PTC	629.786	768.348	3,193.905	112.235	-83.776	-0.021	NAUO9
PTC-03	PTC	630.052	2,116.226	1,849.222	1,456.918	1,264.102	0.245	NAUO7
R7-TC-A	PTC	1,254.296	4,445.976	3,032.898	273.242	3,593.852	624.489	NAUO10
R7-TC-B	PTC	760.539	1,287.294	167.758	3,138.382	435.169	130.733	NAUO8
R7-TC-extra	unassigned_TC	663.374	598.813	1,845.730	1,460.410	-253.311	33.567	NAUO24/NAUO34
STC-01	STC	646.507	773.296	3,221.810	84.330	-78.828	16.700	NAUO12
STC-02	STC	646.491	980.214	3,305.412	0.728	128.090	16.685	NAUO15
STC-03	STC	645.778	1,448.311	3,311.022	-4.882	596.187	15.971	NAUO14
STC-04	STC	646.044	1,849.189	3,308.757	-2.617	997.065	16.237	NAUO13
STC-05	STC	646.470	2,304.918	3,304.508	1.632	1,452.794	16.663	NAUO16
STC-06	STC	646.505	2,348.058	3,162.153	143.987	1,495.933	16.698	NAUO17
STC-07	STC	646.778	2,102.534	1,772.497	1,533.643	1,250.410	16.971	NAUO18
STC-08	STC	646.711	1,870.465	1,678.920	1,627.220	1,018.340	16.904	NAUO19
STC-09	STC	645.704	1,457.358	1,675.065	1,631.075	605.233	15.897	NAUO20
STC-10	STC	646.552	1,001.580	1,681.101	1,625.039	149.456	16.746	NAUO21
STC-11	STC	646.248	618.485	1,676.893	1,629.247	-233.639	16.441	NAUO22
STC-12	STC	646.507	524.330	1,809.363	1,496.777	-327.794	16.700	NAUO23

STC = pipe-surface reference; LF/PTC/JTC = mount-body centroid; R7-TC-extra = modeled shaft midpoint. Sensing junctions are not encoded. Full paths, source filenames, reference basis and distance diagnostics are in sensors.csv.

REV8 sensor coordinate schedule

Direct STEP extraction | Coordinates and lengths in mm

ID	Type	CAD X	CAD Y	CAD Z	Drawing x	Height y	Normal	STEP path below root
JTC-A	PTC	630.458	2,397.359	2,912.585	393.555	1,545.235	0.652	NAUO9
JTC-B	PTC	628.995	2,053.093	2,044.757	1,261.383	1,200.969	-0.812	NAUO22
LF-01	LF	629.807	2,237.407	3,306.140	0.000	1,385.283	0.000	NAUO1/NAUO46
LF-02	LF	629.786	768.348	3,193.905	112.235	-83.776	-0.021	NAUO8
LF-03	LF	630.052	2,116.226	1,849.222	1,456.918	1,264.102	0.245	NAUO7
LF-04	LF	629.830	637.197	1,679.936	1,626.204	-214.927	0.023	NAUO2/NAUO46
PTC-01	PTC	629.776	2,336.325	3,097.560	208.580	1,484.201	-0.031	NAUO24
PTC-02	PTC	629.786	768.348	3,193.905	112.235	-83.776	-0.021	NAUO8
PTC-03	PTC	630.052	2,116.226	1,849.222	1,456.918	1,264.102	0.245	NAUO7
PTC-04	PTC	629.805	528.200	1,832.075	1,474.065	-323.924	-0.002	NAUO23
STC-01	STC	646.507	773.296	3,221.810	84.330	-78.828	16.700	NAUO10
STC-02	STC	646.491	980.214	3,305.412	0.728	128.090	16.685	NAUO13
STC-03	STC	645.778	1,448.311	3,311.022	-4.882	596.187	15.971	NAUO12
STC-04	STC	646.044	1,849.189	3,308.757	-2.617	997.065	16.237	NAUO11
STC-05	STC	646.470	2,304.918	3,304.508	1.632	1,452.794	16.663	NAUO14
STC-06	STC	646.505	2,348.058	3,162.153	143.987	1,495.933	16.698	NAUO15
STC-07	STC	646.778	2,102.534	1,772.497	1,533.643	1,250.410	16.971	NAUO16
STC-08	STC	646.711	1,870.465	1,678.920	1,627.220	1,018.340	16.904	NAUO17
STC-09	STC	645.704	1,457.358	1,675.065	1,631.075	605.233	15.897	NAUO18
STC-10	STC	646.552	1,001.580	1,681.101	1,625.039	149.456	16.746	NAUO19
STC-11	STC	646.248	618.485	1,676.893	1,629.247	-233.639	16.441	NAUO20
STC-12	STC	646.507	524.330	1,809.363	1,496.777	-327.794	16.700	NAUO21

STC = pipe-surface reference; LF/PTC/JTC = mount-body centroid; R7-TC-extra = modeled shaft midpoint. Sensing junctions are not encoded. Full paths, source filenames, reference basis and distance diagnostics are in sensors.csv.

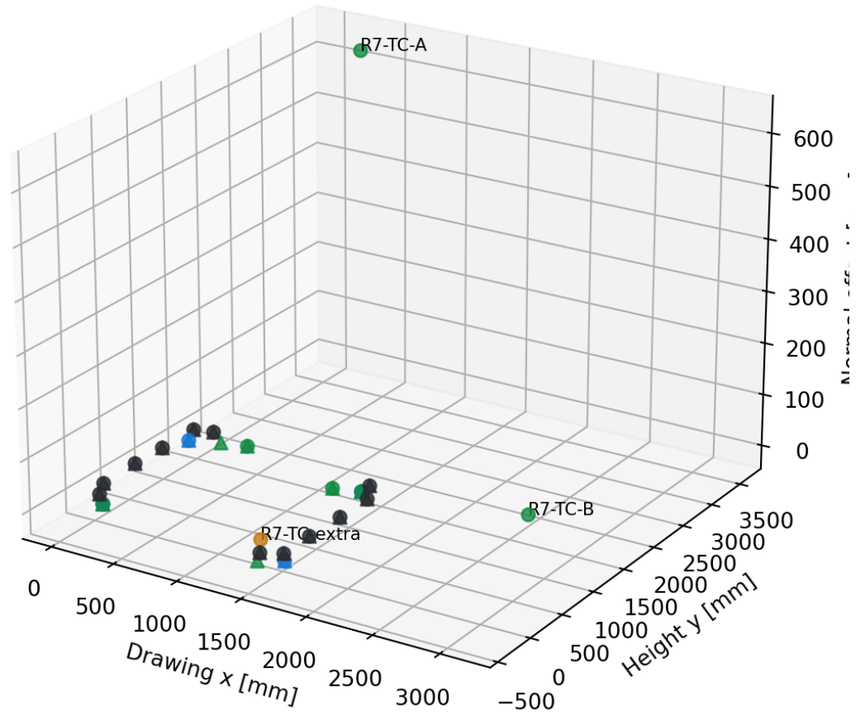
Unmatched probes and out-of-plane locations

Direct STEP extraction | Coordinates and lengths in mm

Revision / ID	CAD X	CAD Y	CAD Z	Main-pipe solid distance
rev7 / R7-TC-B	760.539	1,287.294	167.758	1,501.107
rev7 / R7-TC-A	1,254.296	4,445.976	3,032.898	2,168.242
rev7 / R7-TC-extra	663.374	598.813	1,845.730	-
rev8 / PTC-04	629.805	528.200	1,832.075	0.000
rev8 / PTC-01	629.776	2,336.325	3,097.560	0.000

R7-TC-A/B are remote assembly placements, not established physical installations. Do not subtract them from PTC-01/04 as confirmed sensor motion. All four possible pair displacements are retained separately, clearly marked unconfirmed.

Sensor references in 3D: Rev7 circles, Rev8 triangles



Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

Geometric TC shaft endpoints: all modeled probe shafts

Direct STEP extraction | Coordinates and lengths in mm

Rev / shaft part path below root	Start X	Start Y	Start Z	End X	End Y	End Z	Length
rev7 / NAUO7/NAUO49	710.184	2,242.591	1,826.940	629.459	2,115.290	1,849.386	152.400
rev7 / NAUO8/NAUO37/NAUO39	563.916	1,287.277	167.729	722.666	1,287.277	167.729	158.750
rev7 / NAUO9/NAUO49	692.086	904.110	3,169.966	629.325	767.343	3,194.082	152.400
rev7 / NAUO10/NAUO37/NAUO39	1,313.996	4,261.479	3,065.414	1,265.773	4,410.430	3,039.150	158.750
rev7 / NAUO11/NAUO37/NAUO39	826.767	2,388.863	2,905.433	668.269	2,395.699	2,911.195	158.750
rev7 / NAUO24/NAUO34	697.464	665.927	1,833.896	629.283	531.699	1,857.564	152.400
rev7 / NAUO25/NAUO37/NAUO39	814.760	2,068.344	2,107.362	664.770	2,056.054	2,056.827	158.750
rev8 / NAUO7/NAUO39	710.184	2,242.591	1,826.940	629.459	2,115.290	1,849.386	152.400
rev8 / NAUO8/NAUO39	692.086	904.110	3,169.966	629.325	767.343	3,194.082	152.400
rev8 / NAUO9/NAUO27/NAUO29	826.767	2,388.863	2,905.433	668.269	2,395.699	2,911.195	158.750
rev8 / NAUO22/NAUO27/NAUO29	814.760	2,068.344	2,107.362	664.770	2,056.054	2,056.827	158.750
rev8 / NAUO23/NAUO27/NAUO29	636.235	720.682	1,792.453	631.027	565.271	1,824.422	158.750
rev8 / NAUO24/NAUO27/NAUO29	720.176	2,507.221	3,061.745	647.174	2,369.245	3,090.640	158.750

These are axis endpoints of the longest 3.175 mm-diameter cylindrical shaft in each TC part. Start/end follow the CAD parameterization; neither endpoint is automatically a thermoelectric junction. Parts are associated with mounts through their complete assembly paths.

The package also includes all 1,962 cylindrical-face records within the sensor hardware, including fiber tubes and fittings. This keeps geometric evidence available without interpreting every fitting as a separate sensor.

Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

Exit piping, cooling jacket and expansion tank

Direct STEP extraction | Coordinates and lengths in mm

Rev / component	Start X	Start Y	Start Z	End X	End Y	End Z	Length	OD
rev7 / TLEP	629.807	2,366.318	3,234.247	629.807	2,836.218	3,234.247	469.900	33.401
rev7 / expansion_tank	629.807	2,835.964	3,234.247	629.807	3,147.114	3,234.247	311.150	203.200
rev7 / BREP	629.726	45.564	1,751.829	629.807	515.464	1,751.829	469.900	33.401
rev7 / cooling_jacket	629.807	2,138.709	1,976.487	629.807	2,315.411	2,978.615	1,017.587	88.900
rev8 / TLEP	629.807	2,366.318	3,234.247	629.807	2,836.218	3,234.247	469.900	33.401
rev8 / expansion_tank	629.807	2,835.964	3,234.247	629.807	3,147.114	3,234.247	311.150	203.200
rev8 / BREP	629.726	45.564	1,751.829	629.807	515.464	1,751.829	469.900	33.401
rev8 / cooling_jacket	629.807	2,138.709	1,976.487	629.807	2,315.411	2,978.615	1,017.587	88.900

Axes are extracted from the outer cylindrical faces, with endpoints ordered from lower to higher CAD Y. TLEP is the top-left exit pipe; BREP is the bottom-right exit pipe. Jacket dimensions refer to the modeled shell, not an inferred active heat-transfer length.

Component	Signed dx	Signed dy	Length	Comparison
TLEP	-0.000	469.900	469.900	Coordinates agree between revisions
expansion_tank	-0.000	311.150	311.150	Coordinates agree between revisions
BREP	-0.000	469.900	469.900	Coordinates agree between revisions
cooling_jacket	-1,002.128	176.702	1,017.587	Coordinates agree between revisions

Sources: UTDTloopWithOpticsAndHXRev7.STEP and UTDTloopWithOpticsAndHXRev8(1).STEP - CAD extraction

Method, source provenance and verification

Direct STEP extraction | Coordinates and lengths in mm

Direct extraction. Both attached STEP files were opened with Open CASCADE via CadQuery/XCAF. All parent assembly transforms were composed recursively. STEP units are converted to the importer millimetre coordinate system. No prior workbook coordinates or PDF measurements enter the calculation.

Corner construction. Main-pipe outer cylindrical faces define the straight-axis endpoints. Duplicate face representations are deduplicated for length sums. Small connector stubs remain in the raw axis table. Adjacent cooler pieces are grouped as one named leg while retaining the measured axis gap.

Independent endpoint check. Each STEP file was reloaded and boundary vertices were projected onto the twelve long main-pipe cylindrical faces. The endpoints agree with the analytical cylinder-parameter bounds to less than 1e-6 mm (maximum observed 9.1e-13 mm). This check uses the CAD again, not the prior table.

Sensor completeness. Rev7: 12 wall contacts, four LF references, six TC mount references and one additional TC shaft midpoint (23 records). Rev8: 12 wall contacts, four LF references and six TC mount references (22 records). Shared LF/PTC mounts count as two instrument references at one station.

Reference uncertainty. Surface contact is a nearest-point solution, not a surveyed installation. LF spatial sensing extent and PTC sensing junction locations are not defined by holder centroids. Two TC mounts appear associated with cooling-jacket ports. The inferred service and schema labels remain distinct from CAD facts.

Reproduction. The ZIP contains full-precision CSV/JSON, raw recursive extraction, endpoint verification results, PNG/SVG plots and the Python extraction/analysis scripts. Source STEP files are identified by the hashes below. The original uploaded CADs were not modified.

UTDTloopWithOpticsAndHXRev7.STEP
SHA-256: c07b6be2449d8b5834ce37d53e1550b35a657c33e17a4c9e10442d76e56ebbea

UTDTloopWithOpticsAndHXRev8(1).STEP
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